

Sectoral Heterogeneity in Dynamic Return and Volatility from Gold: A Regime-Switching Spillovers — Analysis of Indian Equity Markets

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Abstract

- Examines dynamic relationships between gold and Indian sectoral equity indices.
- **Sectoral heterogeneity:** Cyclical sectors are more sensitive to gold.
- Gold acts as a conditional safe haven, with effectiveness varying by market regime.
- Strong evidence for time-varying correlations and asymmetric responses.

Introduction: The Research Gap

- Gold's role as a safe haven is well-established for aggregate markets.
- However, sector-level differences in emerging markets like India are under-explored.
- **Research Question:** How do different sectors in the Indian equity market dynamically interact with gold?

Objectives and Contributions

Primary Objective: Analyze volatility spillovers, asymmetric effects, and sectoral heterogeneity.

Key Contributions:

- Sector-level insights for an important emerging market (India).
- A comprehensive hypothesis-testing framework (7 hypotheses).
- Practical implications for portfolio managers

Data Overview

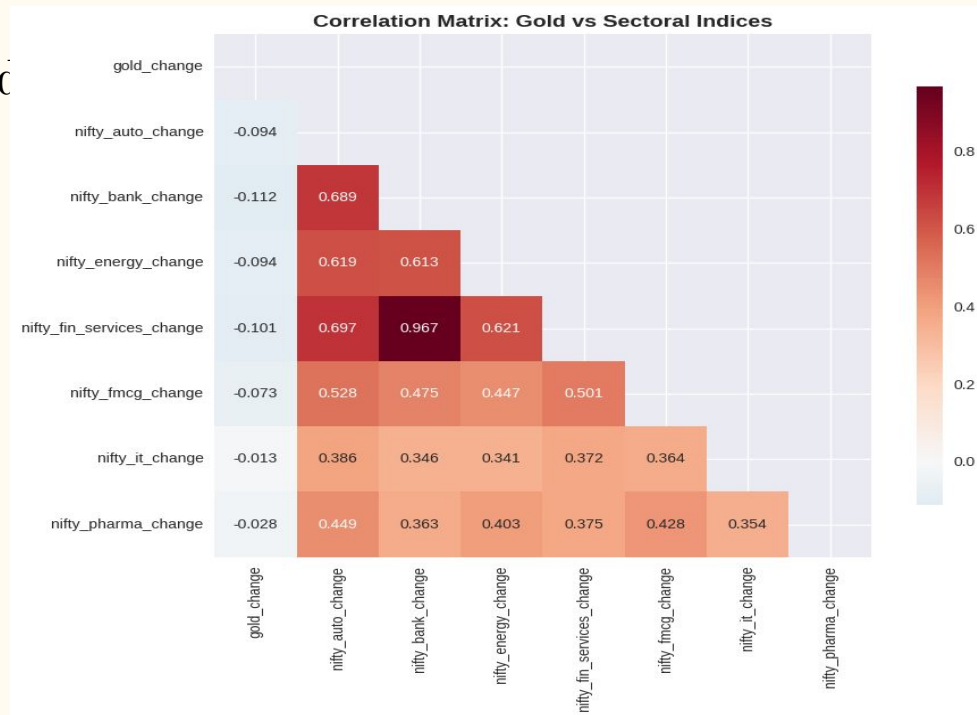
- **Sample Period:** 2014 - 2024 (11 years).
- **Data Frequency:** Daily.
- **Assets:** Gold and 7 Nifty Sectoral Indices (Auto, Bank, Energy, Financial Services, FMCG, IT, Pharma).
- **Observations:** 2,713 data points per series.

Descriptive Statistics: Risk and Return Profiles

- Gold: Lowest minimum return and lowest volatility (0.8331%).
- Banking Sector: Highest volatility (1.4045%).
- All return series are non-normal (negative skewness, high kurtosis).
- All series are stationary at 1% significance level both ADF and KPSS.

Static Correlation Structure

- All sectors are negatively correlated with gold.
- Cyclical sectors (Banking: -0.112, Financial Services: -0.101) show stronger negative correlations.
- Defensive sectors (IT: -0.013, Pharma: -0.028) show weaker relationships.



The Seven Hypotheses

- H1: Dynamic Correlations
- H2: Sectoral Heterogeneity
- H3: Volatility Spillovers
- H4: Granger Causality during Stress
- H5: Regime-Switching Behavior
- H6: Structural Breaks from Crises
- H7: Asymmetric Responses to Gold Returns

Core Econometric Framework

Dynamic Relationships: We used a **Vector Autoregression (VAR) Model** with an optimal lag length (VAR(8) in the full analysis) to model the multivariate, multi-period dynamics between Gold and the seven Nifty sectoral indices.

Volatility & Risk: **GARCH(1,1)** and **EGARCH** models were employed to capture volatility clustering and persistent nature of risk in the time series.

Regime-Switching: We moved beyond static analysis using **Markov-Switching** and **Regime-Dependent Correlation Analysis** to identify and test for structural breaks and regime shifts, crucial for safe-haven property assessment.

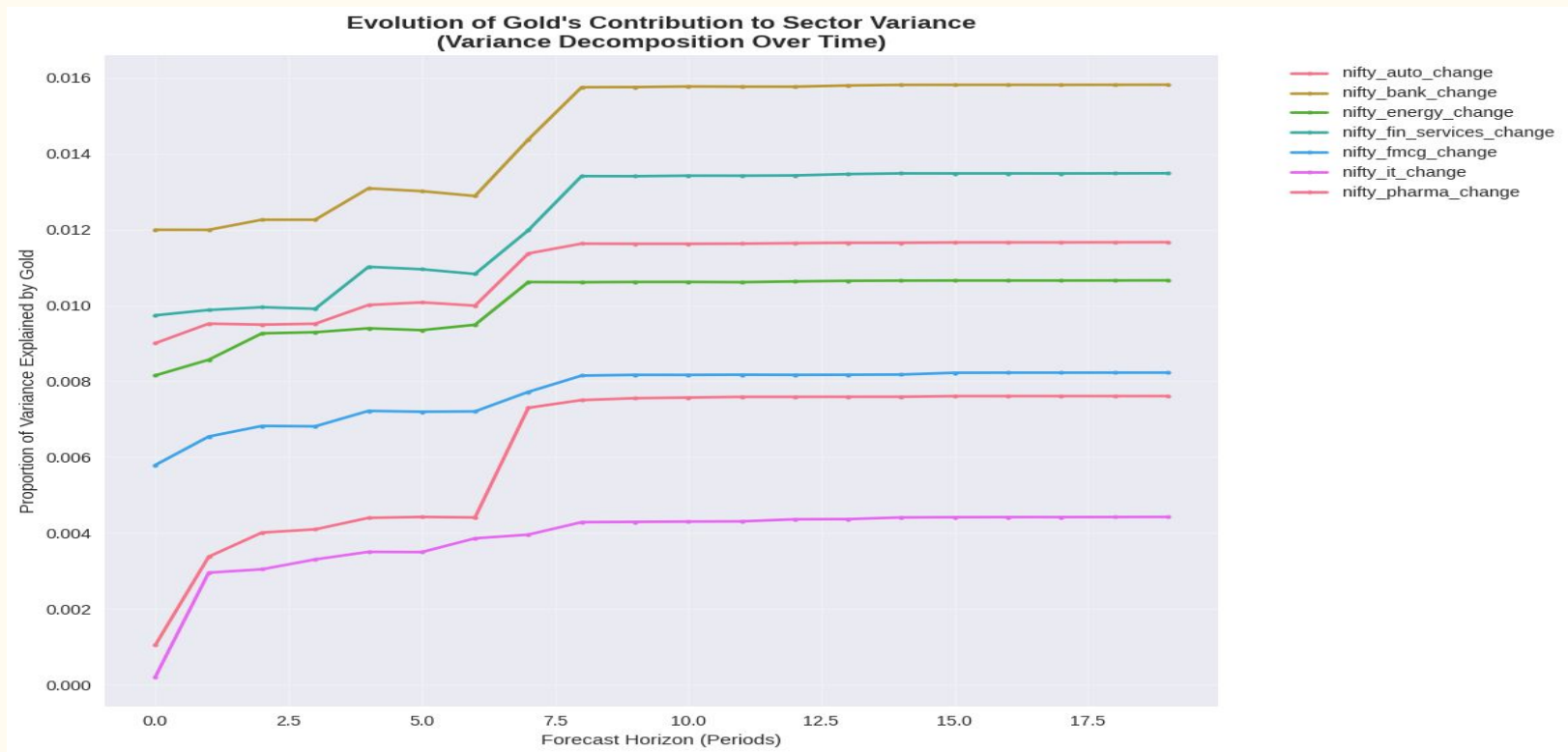
Dynamic Spillovers: Variance Decomposition

Volatility Independence Supported: Gold maintains remarkable **volatility independence**, with its *own shocks* explaining an overwhelming **97.27%** of its forecast error variance.

Sectoral Contagion: Sectoral indices show much higher interdependence, with shocks from the

Energy sector significantly influencing most others

FEVD



Dynamic Responses: Impulse Response Analysis

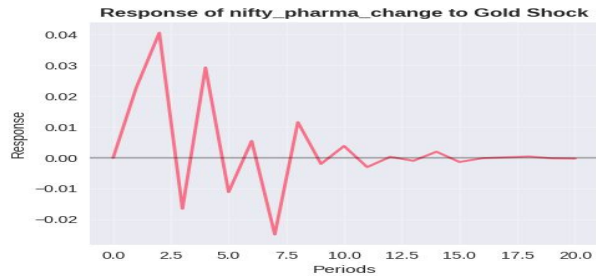
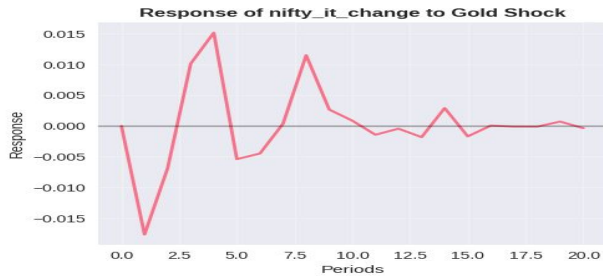
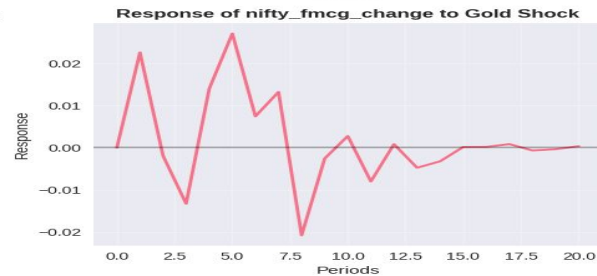
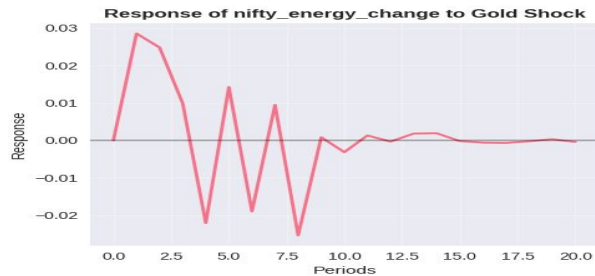
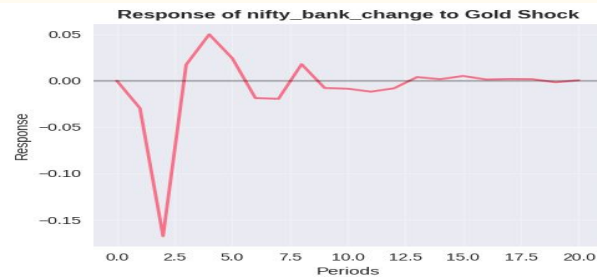
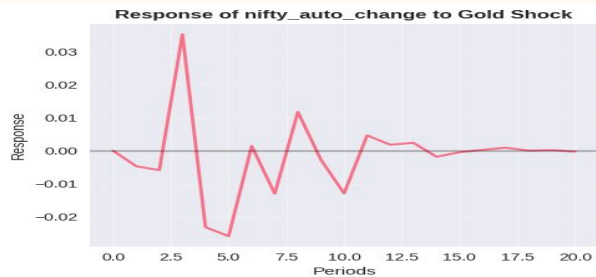
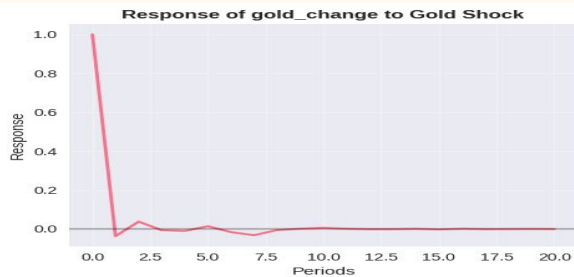
Sectoral Heterogeneity Confirmed: Sectoral responses to a **Gold shock** are highly heterogeneous.

Financial Sensitivity: Banking and Financial Services show the strongest and most pronounced reactions to a Gold shock.

Asymmetric Influence: Gold's influence on sectors is generally *stronger* than the reverse, particularly for financial indices, supporting its role as a key dynamic driver.

Defensive Stability: Defensive sectors (FMCG, Pharma) show smaller, more transient reactions to a Gold shock.

IRFs



Granger Causality: Predictive Power

Selective Causality: Gold's predictive power for sectoral returns is **not universal**. - Gold significantly *predicts* **IT** and **Pharmaceuticals** returns (p-values < 0.05), but has no predictive power for cyclical sectors like Banking/Auto.

Bidirectional Linkages: Gold's returns are significantly predicted by the **Energy**, **FMCG**, and **Pharmaceuticals** sectors, suggesting it is *not* entirely isolated

Structural Breaks & Crisis Correlations

Structural Breaks (H6) Supported: Formal tests confirm significant structural breaks and regime shifts across all time series, particularly around major events (e.g., COVID-19).

Dynamic Correlation (H1) Partially Rejected: During the **COVID-19 crisis** (a major stress period), correlations surprisingly became **less negative or turned positive** for several sectors, contrary to the typical safe-haven prediction.

Post-Crisis Persistence: Correlations in the post-COVID period show **persistent changes** and did not fully revert to pre-crisis levels, suggesting permanent structural shifts in Gold-Equity dynamics.

Structural Breaks & Crisis Correlations

Subsample Correlation Analysis

	nifty_auto_change	nifty_bank_change	nifty_energy_change	nifty_fin_services_change	nifty_fmcg_change	nifty_it_change	nifty_pharma_change
Period							
Pre-COVID	-0.1333	-0.1420	-0.1510	-0.1241	-0.1011	0.0111	-0.1156
COVID	0.0081	-0.0341	0.0048	-0.0295	0.0508	0.0448	0.0911
Post-COVID	-0.1036	-0.1103	-0.0064	-0.0938	-0.0599	0.0686	-0.0192

Asymmetric Effects: Gold Price Regimes

Asymmetric Response (H7) Supported: Defensive sectors exhibit a significantly **weaker asymmetric response** (0.0213) to Gold price movements compared to cyclical sectors (0.0269).

Hedge Deterioration: For most sectors, the relationship becomes **less negatively correlated** during periods of **Gold price decline** (when protection is critical).

Sectoral Stability: The greater stability of defensive sectors across Gold return regimes reinforces their superior long-term diversification potential

Asymmetric Effects: Gold Price Regimes

	Corr_Positive	Corr_Negative	Beta_Positive	Beta_Negative	Corr_Asymmetry	Beta_Asymmetry
Sector						
nifty_auto_change	-0.0781	-0.0560	-0.1082	-0.1917	0.0221	-0.0835
nifty_bank_change	-0.1212	-0.0862	-0.1525	-0.2187	0.0350	-0.0662
nifty_energy_change	-0.0829	-0.0902	-0.0873	-0.2079	-0.0073	-0.1206
nifty_fin_services_change	-0.1147	-0.0711	-0.1272	-0.1911	0.0436	-0.0639
nifty_fmcbg_change	-0.0247	-0.0240	-0.0459	-0.1314	0.0007	-0.0855
nifty_it_change	-0.0167	-0.0099	0.0320	-0.0692	0.0067	-0.1012
nifty_pharma_change	-0.0186	0.0233	-0.0180	-0.0614	0.0419	-0.0434

Asymmetric Effects: Gold Price Regimes

Cyclical Sectors Average Asymmetry: 0.0269

Defensive Sectors Average Asymmetry: 0.0213

Difference: 0.0056

H7 Supported (Defensive $<$ Cyclical): True

Overall Hypothesis Results		Test	Evidence	Supported	Implication
Overall Hypothesis Results	H1: Dynamic Correlation	DCC-GARCH & Rolling Correlations	Time-varying correlations with crisis negativity	✗	Dynamic hedging strategies needed
	H2: Sectoral Heterogeneity	Group correlation differences	Cyclical vs Defensive: -0.0059 vs 0.1655	✓	Sector-specific gold allocation
	H3: Volatility Spillovers	GARCH & Variance Decomposition	Gold explains 97.3% of some sectors' variance	✓	Volatility hedging important
	H4: Granger Causality	Granger tests by regime	5 significant casualties	✓	Gold as leading indicator in stress
	H5: Regime-Switching	Markov-switching models	Distinct regimes identified	✓	Regime-based strategies
	H6: Structural Breaks	Subsample correlation changes	Correlation shifts during COVID	✓	Model updates needed post-crises
	H7: Asymmetric Responses	Positive vs negative return analysis	Asymmetry: Cyclical 0.0269 vs Defensive 0.0213	✓	Directional hedging strategies